

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓	✓		✓
CO 3	✓			✓
CO 4	✓	✓		✓
CO 5		✓		✓
CO 6	✓			

Programme	BSc Statistics
Course Code	STA2MN107 (P)
Course Title	Statistical Inference I
Type of Course	Minor

Semester	II				
Academic Level	100 - 199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Understanding of fundamental statistical concepts. Basic knowledge in probability theory and random Variables.				
Course Summary	To equip students with a comprehensive understanding of theoretical distributions, sampling distributions,, hypothesis testing, and comparisons between independent and paired samples.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Develop a basic understanding of Theoretical Distributions such as Binomial,Poisson and Normal distributions	U	C	Instructor-created exams / Quiz
CO2	Grasp the fundamental principles underlying sampling distributions.	R	C	Practical Assignment / Observation of Practical Skills/ Instructor-created exams
CO3	Analyze and interpret data effectively by comparing two means, calculating confidence intervals for the difference of two population means and analyze data to help entrepreneurial decisions using critical thinking skills.	Ap	F	Seminar Presentation / Group Tutorial Work/ Instructor-created exams
CO4	Evaluate the significance of differences between paired observations.	U	C	Instructor-created exams / Home Assignments
CO5	Explain the principles of relationships between categorical variables, including independence and association and critically evaluate ethical implications of statistical methods aligning with human values.	U	C	One Minute Reflection Writing assignments/ Instructor-created exams
CO6	Demonstrate how to plot probability curves using any software.	Ap	P	Viva Voce/ Instructor-created exams
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

COURSE CONTENT

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	Theoretical Distributions		12	15
	1	Binomial Distribution	3	
	2	Poisson Distribution	3	
	3	Normal Distribution	6	
	Sections from References: Unit 1:14.2 [Ref 1] Unit 2:14.3[Ref 1] Unit 3:14.4 [Ref 1]			
II		Testing of Hypothesis	13	20
	4	Statistical Hypothesis-Simple and Composite, Null and Alternative	1	
	5	Types of errors in testing ,Level of Significance, Critical Region	3	
	6	One tailed and two tailed, p- value	1	
	7	Procedure of testing of hypothesis	1	
	8	Test for Single Proportion-Large Sample	1	
	9	Test of Significance for Difference of Proportions-Large Sample	2	
	10	Test of Significance for a single mean	2	
	11	Test of Significance for difference of Means	2	
	Sections from References: Unit 4: 16.6,16.6.1,16.6.3,16.6.4[Ref 1] Unit 5: 16.6.5,16.6.6,16.6.7[Ref 1] Unit 6: 16.6.8,16.6.9[Ref 1] Unit 7: 16.7[Ref 1] Unit 8: 17.2.1[Ref 1] Unit 9: 17.2.2[Ref 1] Unit 10: 17.3.1[Ref 1] Unit 11: 17.3.2[Ref 1]			
III		Chi- Square Test	11	20

	12	Chi-square Distribution	2	
	13	Chi- Square Test of goodness of fit	2	
	14	Chi Square Test for Independence of Attributes	2	
	15	Degrees of Freedom	1	
	16	2×2 Contingency table	2	
	17	2×k Contingency table	2	
	Sections from References: Unit 12: 18.2[Ref 1] Unit 13: 18.4&18.5[Ref 1] Unit 14: 18.6[Ref 1] Unit 15: 18.7[Ref 1] Unit 16: 18.8[Ref 1] Unit 17: 18.9[Ref 1]			
IV		Small sample Tests	9	15
	18	Student's t distribution	2	
	19	Applications of t distribution	1	
	20	Test for single mean	2	
	21	t- Test for Difference of Means	2	
	22	Paired t- Test for difference of Means	2	
	Sections from References: Unit 18: 19.2[Ref 1] Unit 19: 19.4[Ref 1] Unit 20: 19.5[Ref 1] Unit 21: 19.6[Ref 1] Unit 22: 19.7[Ref 1]			
V	PRACTICUM		30	
	Do practice problems using spreadsheet from any 5 units of the given list and one additional problem decided by the teacher-in-charge, related to the content of the course. Other units listed here may be used as demonstrations of the concepts taught in the course.			
	1	Draw probability histogram		
	2	Finding mean and variance of a probability distribution		
	3	Methods for finding Binomial probability		
	4	Methods for finding Poisson probability		

	5	Find normal distribution areas		
	6	Find z scores from known area of normal probability		
	7	Assessing normality		
	8	Normal quantile plots		
	Sections from References: Unit 1: 5.1 [Ref 5] Unit 2: 5.1 [Ref 5] Unit 3: 5.2 [Ref 5] Unit 4: 5.4 [Ref 5] Unit 5: 6.1 [Ref 5] Unit 6: 6.1[Ref 5] Unit 7: 6.5[Ref 5] Unit 8: 6.5 [Ref 5]			
Books and References: 1. Gupta, S. C.. (2015). Fundamentals of Statistics, Himalaya Publishing House 2. Myra L. Samuels, Jeffrey A. Witmer, Andrew A. Schaffner, Statistics for the Life Sciences ,fifth edition (2016),Pearson Education 3. Gupta, S. C. and Kapoor, V. K. (2020). Fundamentals of Mathematical Statistics, 12 th edition, Sulthan Chand, New Delhi 4. Prem S. Mann (2016), Introductory Statistics 9 th Edition, Wiley 5. Mario F Triola, Elementary Statistics using Excel				

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PSO 5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
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CO 1	3	2	-	-	-	2	1	1	-	-	-	-
CO 2	-	-	-	-	-	3	2	2	-	-	-	-
CO 3	-	-	3	-	-	2	2	2	-	-	3	-
CO 4	-	2	-	-	-	-	2	3	-	-	-	-
CO 5	-	-	-	-	-	2	3	1	-	-	-	3
CO 6	-	2	-	2	3	-	2	2	3	3	-	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal	Assignm	Project	End Semester
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